ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications [ANSWER KEY]

Unit 3.01: Topic 3: Geometry, Trigonometry & Voronoi Diagrams

Q1. In the diagram, triangle ABC is shown with coordinates A1, 1, B3, 1, and C3, 5. What is the exact length of the side AC in simplest radical form?

[ ] A: 4
[ ] B: 5
[x] C: 2sqrt{5}
[ ] D: 4sqrt{2}

Solution: Using the distance formula between A1, 1 and C3, 5 : AC = sqrt{3 - 1^2 + 5 - 1^2} = sqrt{2^2 + 4^2} = sqrt{4 + 16} = sqrt{20} = 2sqrt{5}. [M1] [A1]

Q2. In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

[x] A: 9
[ ] B: 10
[ ] C: 15
[ ] D: 22.5

Solution: By the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments created by the bisector on the opposite side. Therefore, frac{AB}{AC} = frac{BD}{DC}. Substituting the given values, we get frac{10}{15} = frac{6}{DC}. Simplifying the left side, frac{2}{3} = frac{6}{DC}. Cross-multiplying, 2 times DC = 3 times 6. 2 times DC = 18. Dividing by 2, DC = frac{18}{2} = 9. [M1] [A1]

Q3. Volume of cone with radius 3, height 4:

[x] A: 12pi
[ ] B: 16pi
[ ] C: 36pi
[ ] D: 48pi

Solution: Applying Geometry & Trigonometry principles, V=frac{1}{3}pi94=12pi. Therefore, the verified correct option is 12pi. [M1] [A1]

Q4. In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

[x] A: 6
[ ] B: 8
[ ] C: 12
[ ] D: 15

Solution: In a right-angled triangle, the altitude to the hypotenuse creates two smaller triangles that are similar to the original triangle and to each other. According to the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. In triangle ABC, AD is the altitude to the hypotenuse BC. Therefore, we have: AD^2 = BD times DC Given that BD = 4 and DC = 9: AD^2 = 4 times 9 AD^2 = 36 AD = sqrt{36} AD = 6. [M1] [A1]

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Q5. In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

[ ] A: 7
[x] B: 9
[ ] C: 10
[ ] D: 12

Solution: According to the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments it divides the opposite side into. Therefore, frac{AB}{AC} = frac{BD}{DC}. Substituting the given values, we have frac{10}{15} = frac{6}{x}, where x is the length of DC. Solving for x: 10x = 15 times 6 10x = 90 x = frac{90}{10} x = 9. [M1] [A1]

Q6. In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

[ ] A: 5 cm
[x] B: 6 cm
[ ] C: 7 cm
[ ] D: 3 cm

Solution: In a right-angled triangle, when an altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are similar to the original triangle and to each other. This leads to the geometric mean theorem. The length of the altitude BD is the geometric mean of the two segments it divides the hypotenuse into AD and DC. Therefore, BD^2 = AD times DC . Given AD = 4 cm and DC = 9 cm, BD^2 = 4 times 9 BD^2 = 36 BD = sqrt{36} BD = 6 cm. [M1] [A1]

Q7. In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.

[ ] A: 10
[ ] B: 6
[x] C: 8
[ ] D: 16

Solution: The triangle ABC is a right-angled triangle with the right angle at C. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse the side opposite the right angle is equal to the sum of the squares of the other two sides. In this case, AB is the hypotenuse. So, we have: AB^2 = AC^2 + BC^2 . We are given AB = 10 and BC = 6 . We need to find AC . Substituting the given values into the equation: 10^2 = AC^2 + 6^2 100 = AC^2 + 36 AC^2 = 100 - 36 AC^2 = 64 AC = sqrt{64} AC = 8 . Therefore, the length of side AC is 8 units. [M1] [A1]

Q8. In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

[ ] A: 5
[x] B: 6
[ ] C: 7
[ ] D: 8

Solution: In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. By the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. Thus, AD^2 = BD times DC. Given BD = 4 and DC = 9, we have AD^2 = 4 times 9 = 36. Taking the square root of both sides, we get AD = sqrt{36} = 6. [M1] [A1]